

Armitage attack on Windows XP

(SCRIPT-KIDDIE ATTACK)

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1. Executive Summary

This report is about using a GUI based script-kiddie attacking tool from Kali Linux called Armitage. In this lab we will learn how to identify vulnerabilities of the victim machine and gain access into the victim machine to perform exploitation attacks such as keylogging, screen capturing, modification of files, privilege escalation with meterpreter. This lab will be carried out in a controlled, sandboxed virtual environment.

The core methodology involved a structured attack chain: initial reconnaissance via an NMAP scan to identify the target, vulnerability analysis using Armitage's automated tools,

and the successful exploitation of the **MS08-067 NetAPI vulnerability**. This exploit provided remote administrative access to the victim machine, enabling post-exploitation activities such as remote desktop surveillance and keylogging through a meterpreter session.

The purpose of this exercise was to demonstrate how automated “script-kiddie” tools can be a significant threat. The offensive perspective was used to inform defensive strategies, emphasizing the critical importance of proactive patch management, principle of least privilege as fundamental security controls.

Lastly, this lab served as a powerful practical lesson: understanding the tools and techniques of attackers and help in gaining the skills to think from attackers’ point-of-view and foster a security analyst mindset.

2. Introduction

2.1 LAB OBJECTIVE AND LEARNING OUTCOME

The objective of this Lab is to configure a virtual lab environment with **Kali Linux (Attacker)** and **Windows XP (Victim)** on an isolated NAT network. We will then execute a simulated attack using the Armitage GUI to exploit the **MS08-067** vulnerability and perform post exploitation activities on the compromised host using Meterpreter.

MS08-067 is a critical remote code execution vulnerability in the server service on Microsoft Windows System. It is one of the most famous and widely exploited vulnerabilities in history. **CVE Identifies: CVE-2008-4250**. This vulnerability could be exploited without needing a username or password.

Script-Kiddie Attack is the attacks carried out by hackers who use pre-written scripts to carryout cyberattacks, these types of hackers lack programming skills and expertise in hacking and use the pre-made tools to launch cyberattacks.

Learning Outcomes

At the end of this lab, I was able to gain practical experience with automated penetration testing tools (Armitage), I gain an insight of post-exploitation techniques and their impact and acquired the knowledge of implementing security measures from an offensive action.

2.2 TOOLS AND TECHNOLOGIES USED

Virtualization Software: Oracle VirtualBox

Attacker OS: Kali Linux 2025

Victim OS: Microsoft Windows XP professional

Penetration Testing Tool: Armitage (GUI for Metasploit)

Payload: Meterpreter

Vulnerability: MS08-067

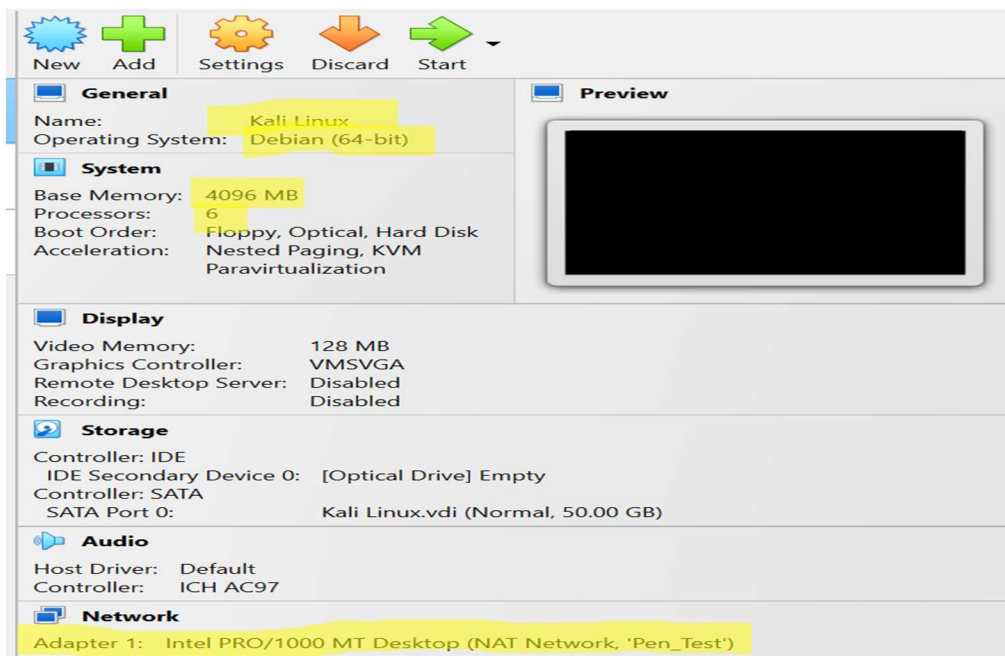
Reconnaissance Tool: Nmap

2.3 DISCLAIMER

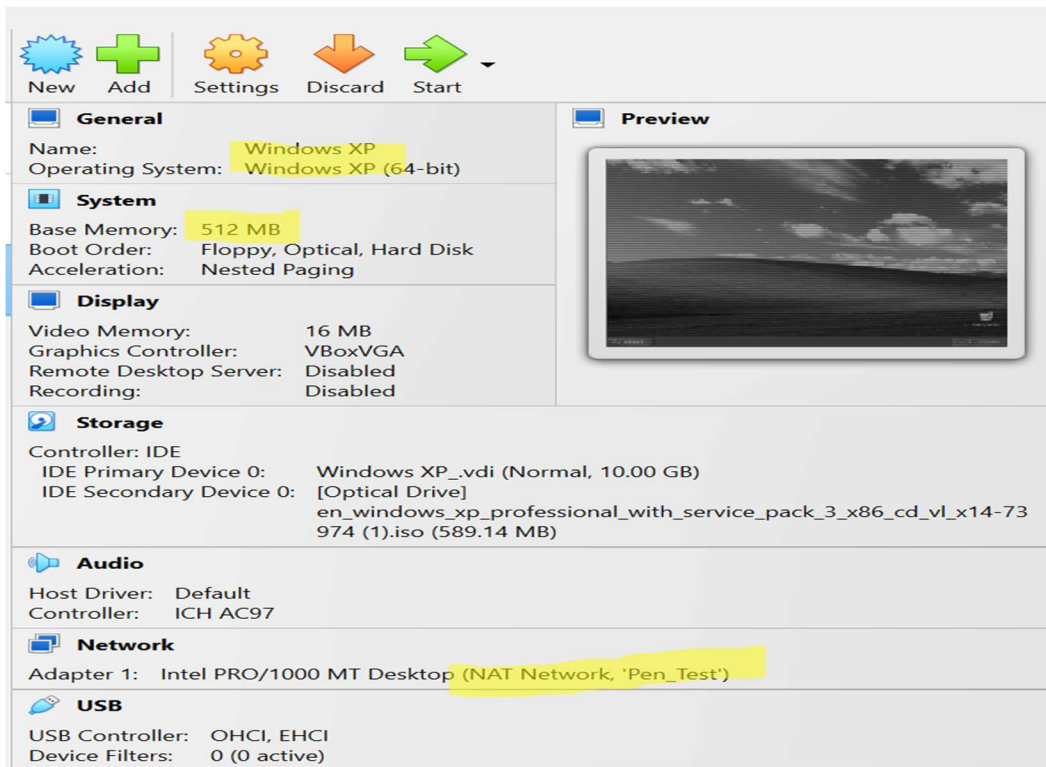
All activities described in this report were performed within a **strictly isolated and self-contained virtual lab environment**. The tools and techniques were used solely for educational purposes to understand attack methodologies and strengthen defensive skills. Unauthorized testing against systems without explicit permission is illegal and unethical.

3. Lab Environment Setup

3.1 KALI LINUX VIRTUAL MACHINE SETUP



3.2 WINDOWS XP VICTIM MACHINE SETUP



3.3 NETWORK CONFIGURATION (NAT NETWORK)

NAT Network acts like a sealed sandbox environment. The traffic between Kali Linux (attacker) and Windows XP (victim) stays entirely within this virtual network ensuring no risk to the real network.

Bridged Adapted: It would put my VMs directly on my home network. It would be dangerous as there could be a risk of attack potentially targeting real devices on my network by accident.

Host-Only: This would isolate the VMs completely, with no internet access. Even though it is safe, it will prevent me from downloading any new tools or updates during the lab session.

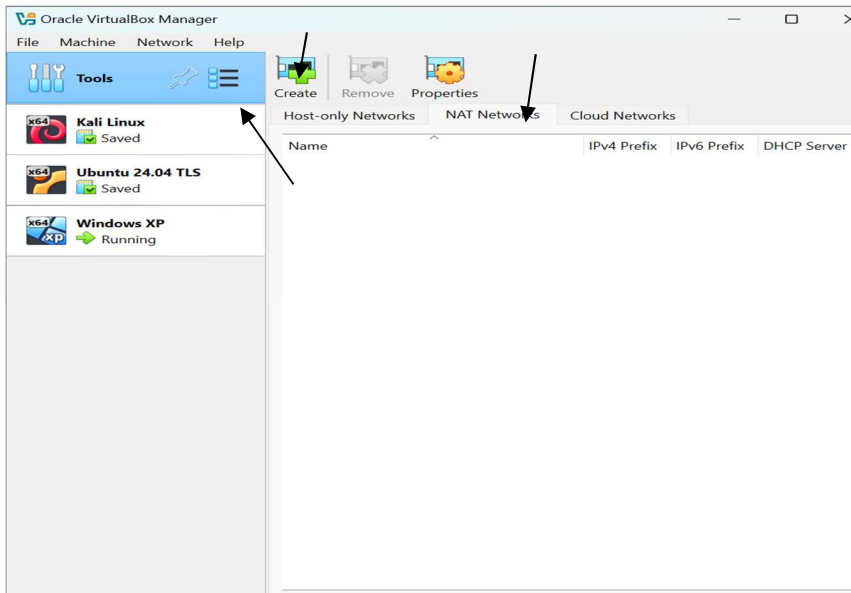
Basic NAT: This mode only provides internet access to a single VM and does not allow VMs to communicate with each other, making the attack impossible.

Therefore, NAT network is the best option for virtual penetration testing environment.

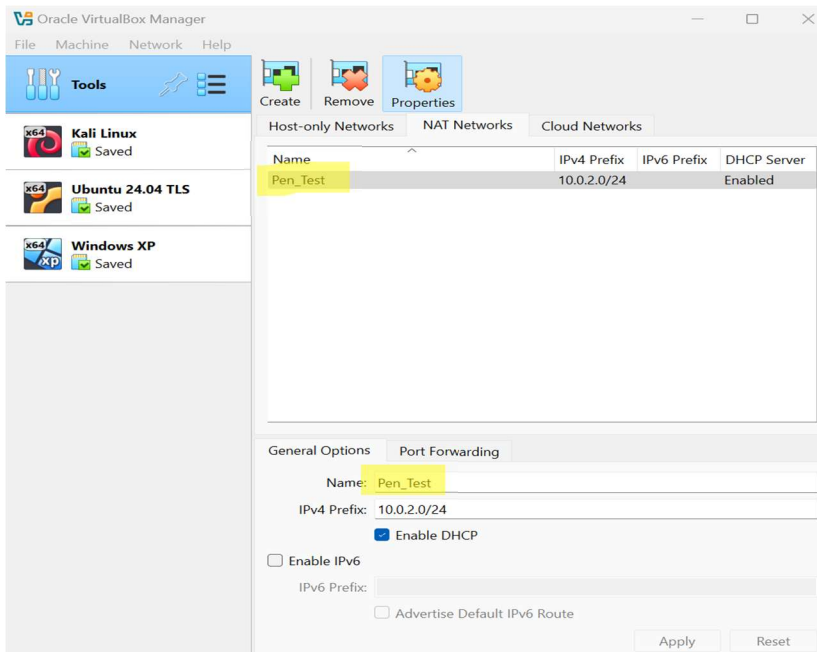
Creating NAT network in VirtualBox

First, we open the VirtualBox Global Settings and then navigate as follow:

Tools → Network → NAT Networks → Create

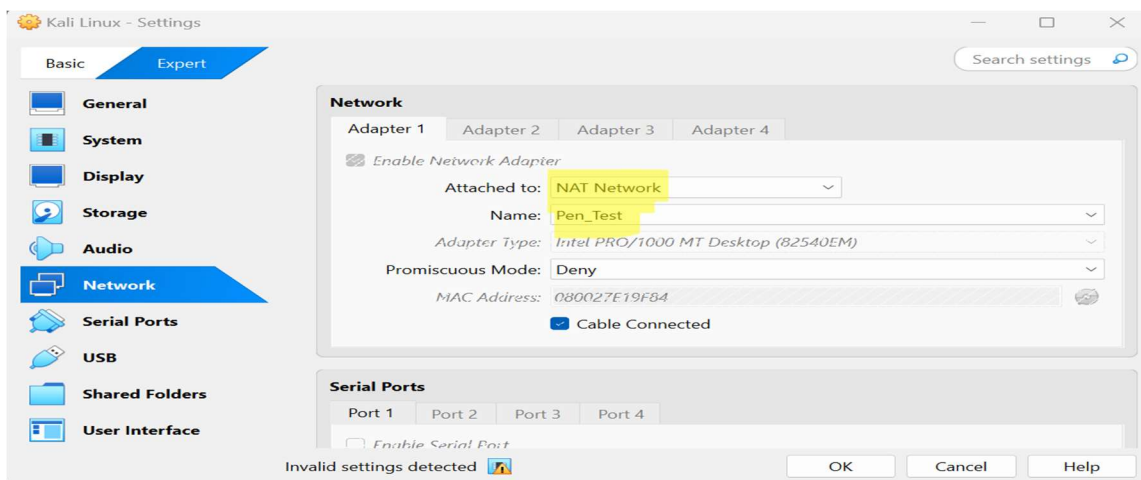


It will create a default NAT network, which we will rename to “Pen_Test”

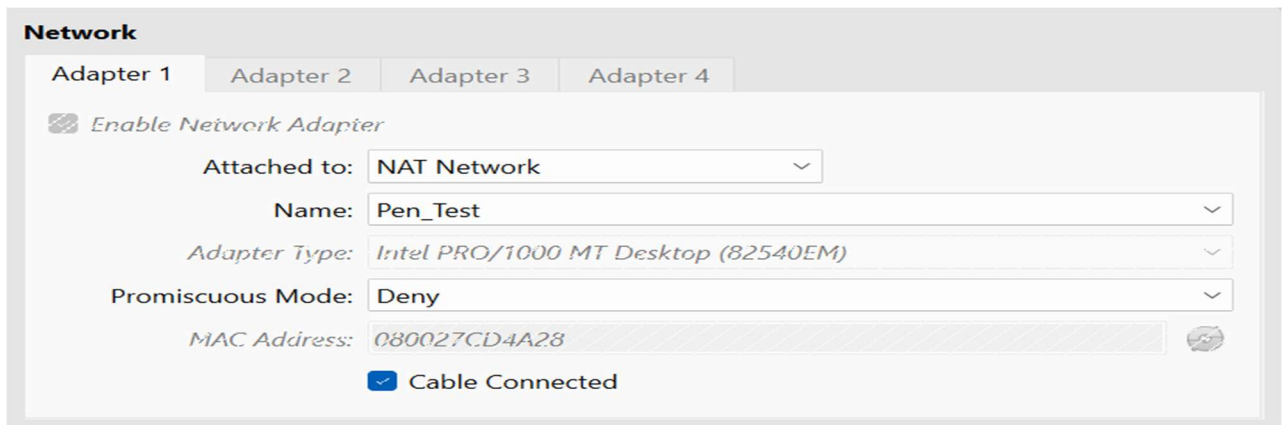


Then we attached our Kali Linux machine and Windows XP to this network, we will do this by going into their settings → network → Attached to (select NAT network).

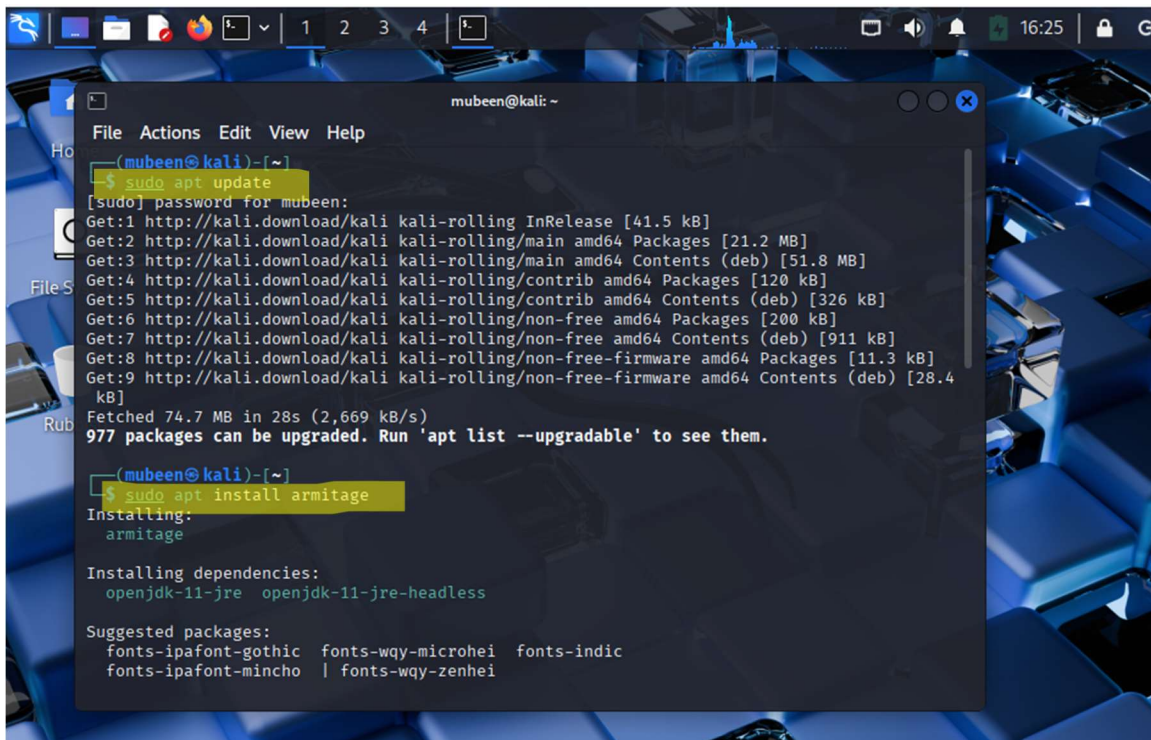
Kali Linux



For Windows XP



Installing Armitage on Kali Linux

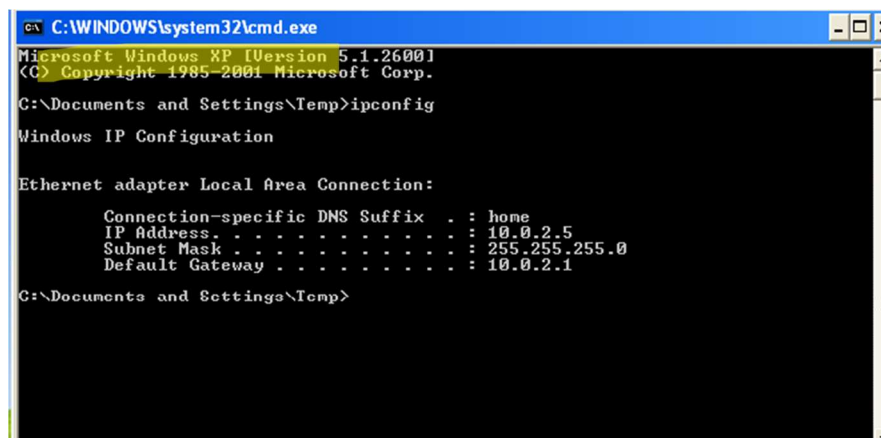


```
mubeen@kali: ~  
File Actions Edit View Help  
(mubeen@kali)-[~]  
$ sudo apt update  
[sudo] password for mubeen:  
Get:1 http://kali.download/kali kali-rolling InRelease [41.5 kB]  
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.2 MB]  
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [51.8 MB]  
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [120 kB]  
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [326 kB]  
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [200 kB]  
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [911 kB]  
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [11.3 kB]  
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [28.4  
kB]  
Fetched 74.7 MB in 28s (2,669 kB/s)  
977 packages can be upgraded. Run 'apt list --upgradable' to see them.  
(mubeen@kali)-[~]  
$ sudo apt install armitage  
Installing:  
armitage  
  
Installing dependencies:  
openjdk-11-jre openjdk-11-jre-headless  
  
Suggested packages:  
fonts-ipafont-gothic fonts-wqy-microhei fonts-indic  
fonts-ipafont-mincho | fonts-wqy-zenhei
```

Finding the Victim and Attacker IP address

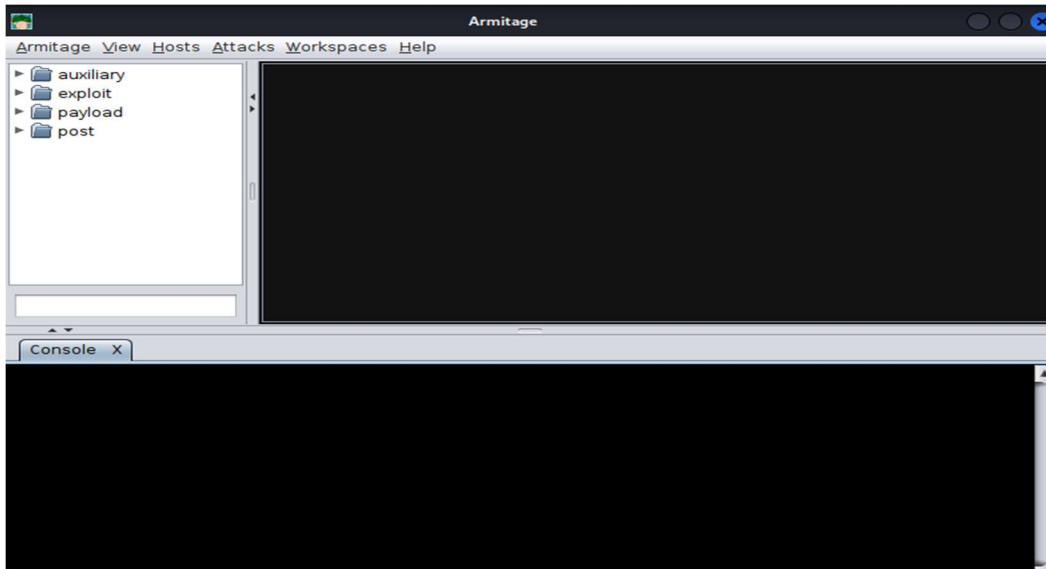
We can use “ipconfig” in the control panel → CMD in windows XP to find the IP address or go to Kali Linux terminal and write “sudo nmap -sn 10.0.2.0/24”

For Kali Linux ip address, I used the command “ip a” in the terminal



```
C:\WINDOWS\system32\cmd.exe  
Microsoft Windows XP [Version 5.1.2600]  
<C> Copyright 1985-2001 Microsoft Corp.  
C:\Documents and Settings\Temp>ipconfig  
Windows IP Configuration  
  
Ethernet adapter Local Area Connection:  
  
Connection-specific DNS Suffix . : home  
IP Address. . . . . : 10.0.2.5  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : 10.0.2.1  
C:\Documents and Settings\Temp>
```

Armitage GUI



4. Methodology: The Attack Chain

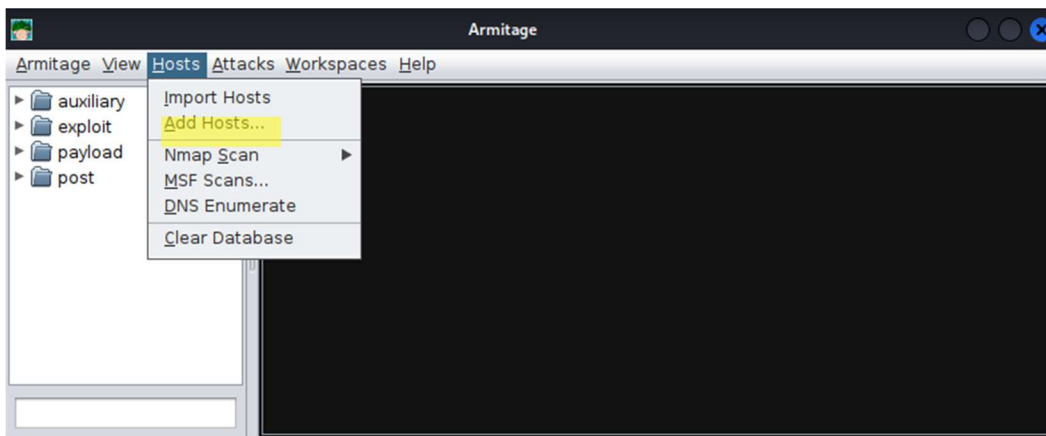
4.1 STEP 1: RECONNAISSANCE

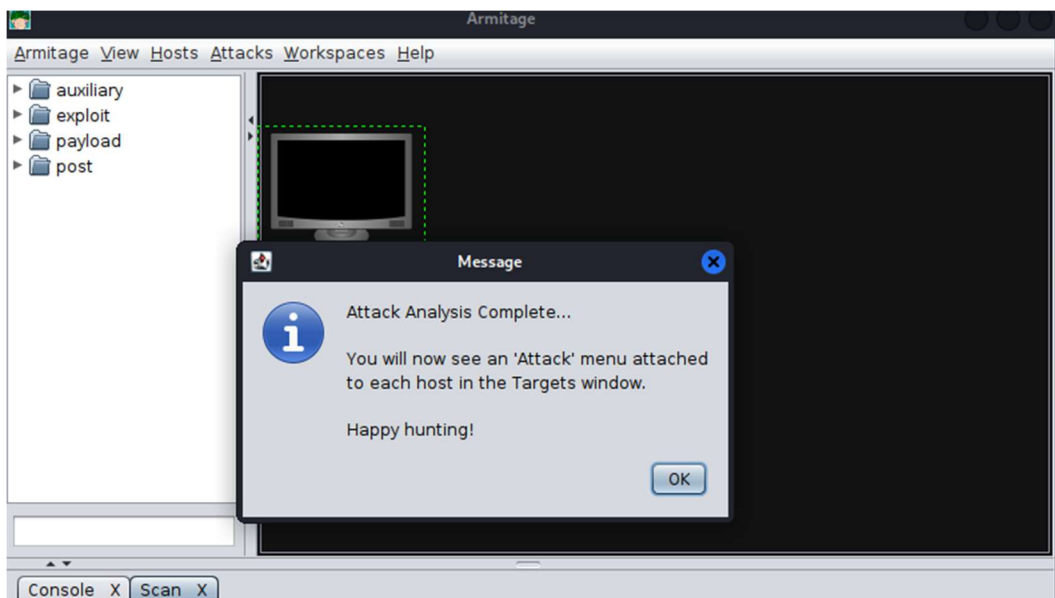
Reconnaissance is the preliminary phase of the cyberattack, it involves scanning of network, systems, or web applications to gather information about potential vulnerabilities that can be exploited.

Tool Used: Integrated NMAP scanner within Armitage GUI

Action:

First, we will go to Host → Add Host and enter the Victim IP address “10.0.2.5”



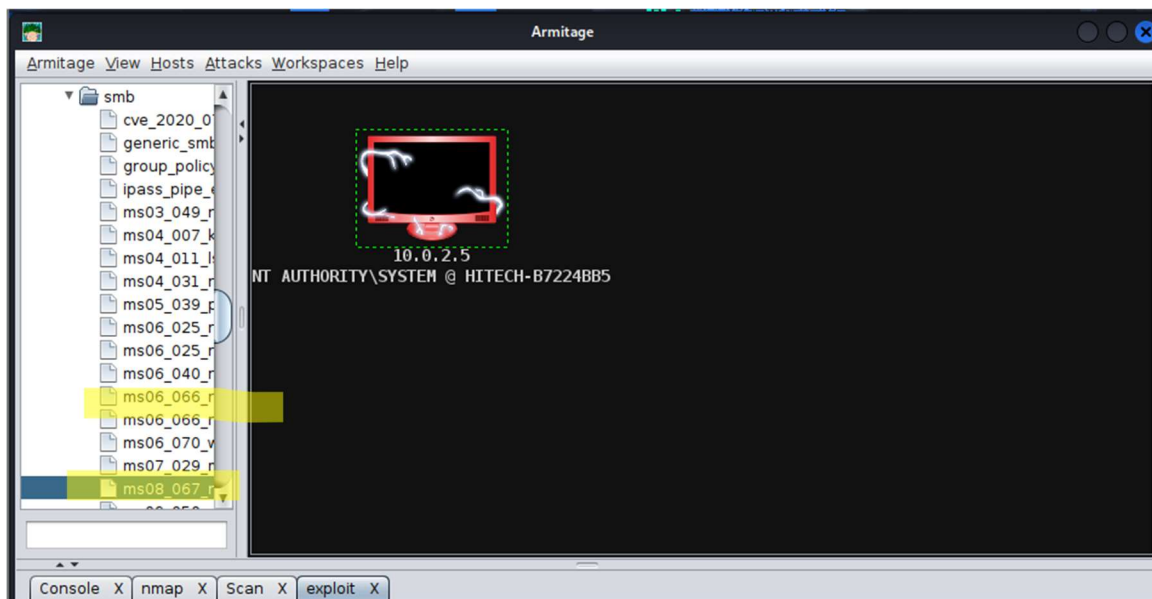


4.3 STEP 3: EXPLOITATION

The exploitation phase was initiated to gain unauthorized access to the target's system.

We start by selecting the exploit on the left-hand panel of Armitage we see exploit.

Navigate as follows: Exploit → Windows → SMB → MSo8_o67NETAPI.



Double Click on MSo8_o67NETAPI

- 1) Set LHOST: 10.0.2.4 (Kali IP)

2) Set RHOST: 10.0.2.5 (Target IP)

Option	Value
LHOST	10.0.2.4
LPORT	7145
RHOSTS +	10.0.2.3
RPORT	445
SMBPIPE	BROWSER

Then click on use reverse connection and launch

Attack 10.0.2.5

MS08-067 Microsoft Server Service Relative Path Stack Corruption

This module exploits a parsing flaw in the path canonicalization code of NetAPI32.dll through the Server Service. This module is capable of bypassing NX on some operating systems and service packs. The correct target must be used to prevent the Server

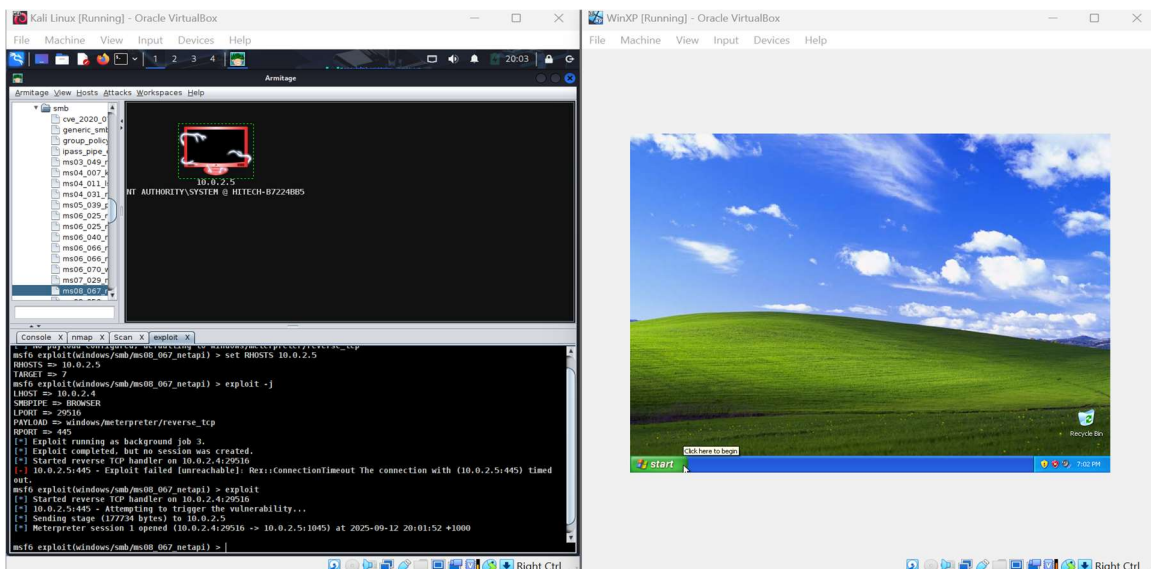
Option	Value
LHOST	10.0.2.4
LPORT	29981
RHOSTS +	10.0.2.5
RPORT	445
SMBPIPE	BROWSER

Targets: 7 => Windows XP SP3 English (NX)

☐ Use a reverse connection

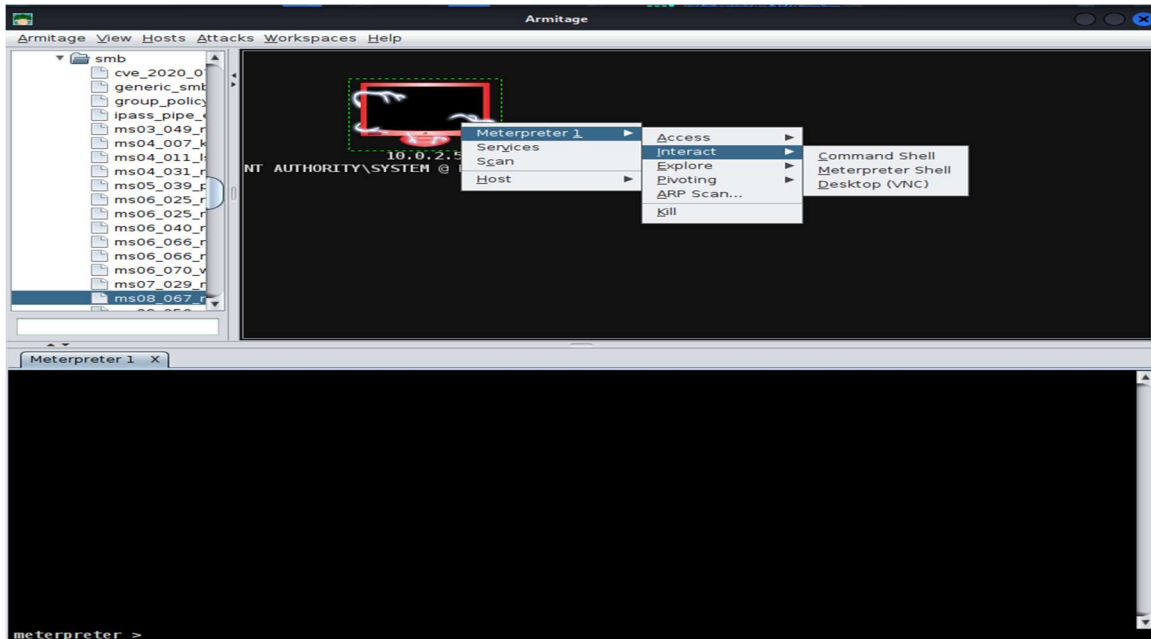
☐ Show advanced options

Launch



We have now successfully triggered the vulnerability on the Windows XP machine. The payload (Meterpreter) was delivered and executed.

A Meterpreter session is now open, and we have the remote-control access to Windows XP. Now we will interact with the session in POST-EXPLOITATION phase.



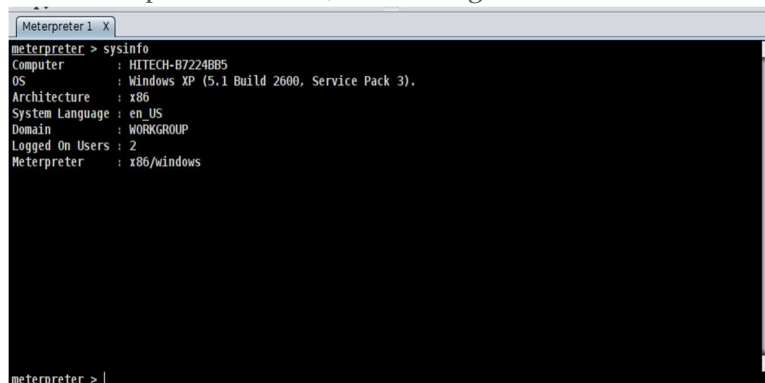
4.4 STEP 4: POST-EXPLOITATION

In this phase, we will demonstrate key post-exploitation techniques.

Activities Performed:

1) System Information Gathering:

We used the command “sysinfo” within the meterpreter session to gather the details about the compromised host, confirming the control over Windows XP system.



2) Remote Desktop Surveillance:

“Screenshot” command was used to capture the current state of the user’s desktop remotely. It demonstrates the ability to visually monitor victim’s computer without their knowledge.

3) Keylogging:

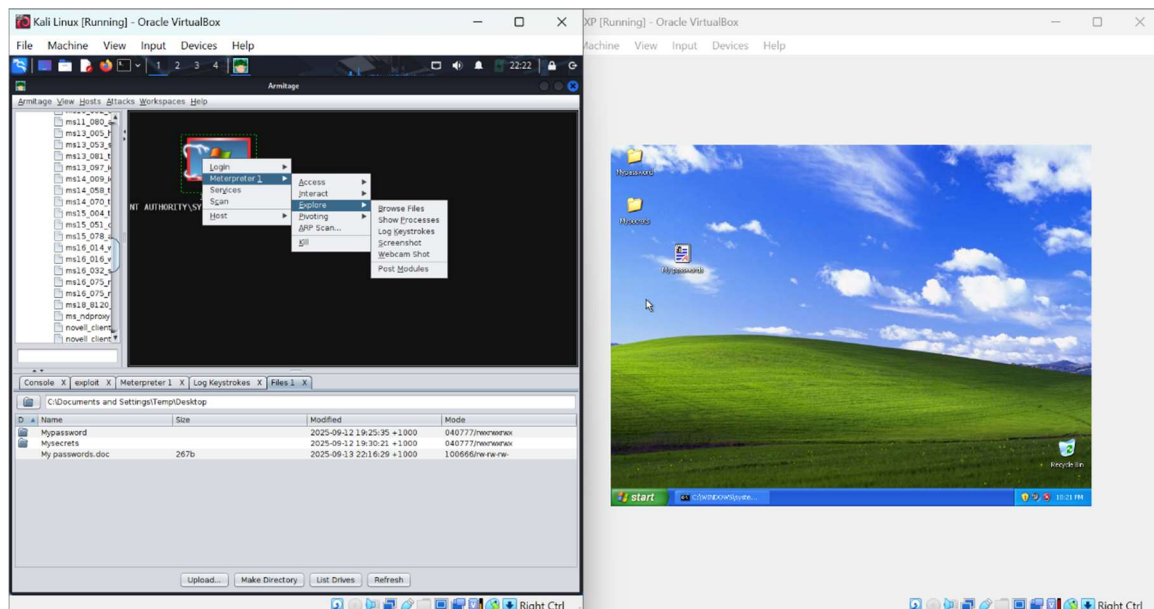
“Keylogging” command was used to begin capturing the keystrokes on the victim machines. Which highlights the risk of stealing sensitive information like passwords and credentials.

```
1 2 Trying CapCore.CAC (1400)
[+] Successfully migrated to Explorer.EXE (1400) as: HITECH-B7224BB5\Temp
[*] Starting the keylog recorder...
[*] Keystrokes being saved in to
/home/mubeen/.msf4/loot/20250913221539_default_10.0.2.5_host.windows.key_392753.txt
[*] Recording keystrokes...
[+] Keystrokes captured <Caps Lock>M<Caps Lock>y passw
[+] Keystrokes captured ords <^H><CR>

[+] Keystrokes captured <Caps Lock>T<Caps Lock>his <Caps Lock>N<Caps Lock>ote
[+] Keystrokes captured pad contains my passwords and us
[+] Keystrokes captured ernes<CR>
<Caps Lock>U<Caps Lock>ername<Shift>: <Caps Lock>M
[+] Keystrokes captured IT233909<CR>
P<Caps Lock>assword<Shift>: <Caps Lock>H<Caps Lock>
[+] Keystrokes captured elloworld123<Shift><Shift><Shift><Shift><Shift><Shift><Shift><Shift>@@
[+] Keystrokes captured <CR>

[+] Keystrokes captured <Left Windows>
msf6 post(windows/capture/keylog_recorder) >
```

Used Browse File option to investigate all the victim files



5. Results & Analysis

The exploitation of MSo8-o67 vulnerability was successful, resulting in a remote Meterpreter session with SYSTEM-level privileges on the Windows XP victim machine. POST-exploitation activities such as keylogging, screenshot capture, shell, file exploration shows the total compromise of confidentiality and control

6. Defense and Mitigation strategies

To defend against exploits like MSo8-o67, organizations must adopt a proactive security posture. Mitigation strategies include:

- Configuring Host-Based Firewalls to block unnecessary inbound traffic to high-risk services from untrusted networks.
- Implement Principle of Least Privilege, restricting users and service account permissions to minimize the impact of a compromise.
- Monitor network traffic for signatures associated with SMB exploitation attempts and block them in real-time.
- Carry out regular vulnerability assessment to identify unpatched systems and misconfigurations proactively.

7. Conclusion

7.1 KEY LEARNINGS

This practical lab developed my foundational skills in offensive security and reinforced critical defensive principles:

- **Tool Proficiency:** Gained hands-on experience with Armitage to automate Metasploit workflows, including NMAP scanning, vulnerability analysis, and exploit execution.
- **Exploit Development:** Understood the process of leveraging a known vulnerability to achieve remote code execution.
- **Post-Exploitation Technique :** Practice essential techniques including privilege escalation, remote desktop surveillance, and user activity monitoring.
- **Troubleshooting:** Developed problem-solving skills by diagnosing network issues, exploit failures- simulating real-world penetration testing challenges.

7.2 ETHICAL HACKER'S MINDSET

The core of ethical hacking is offensive knowledge for defensive purposes. This lab reinforced the mindset that to defend a system, one must first learn how to attack it. This

approach is grounded in a responsibility to protect, requiring strict adherence to legal and ethical boundaries, transparency, and a commitment to using skills to improve security posture rather than to cause harm.